

Total Defaulted Loans:

36639

Default Rate

24.64%

Total # of Loans

148,670

Variable	Missing Values	% Missing	Count of Defaulted Loans (Missing)	Default Rate (Missing Values)	Default Rate (Not Missing)	Notes
rate_of_interest	36439	24.51%	36439	100.00%	0.18%	100% default among missing values. Strong evidence of target leakage/post-outcome recording.
Interest_rate_spread	36639	24.64%	36639	100.00%	0.00%	Missingness perfectly aligns with default. Unsuitable for predictive modeling.
property_value	15098	10.16%	15096	99.99%	16.13%	Missingness almost perfectly predicts default (99.99%). Likely systematic rather than random.
LTV	15098	10.16%	15096	99.99%	16.13%	Strong evidence of target leakage; interpret LTV analyses cautiously.
Upfront_charges	39642	26.66%	36486	92.04%	0.14%	Highly non-random missingness associated with default.
dtir1	24121	16.22%	16310	67.62%	16.32%	Missingness associated with substantially higher default risk. Requires further investigation.
loan_limit	3344	2.25%	881	26.35%	24.61%	Missingness similar to overall portfolio default rate. Limited evidence of bias.
income	9150	6.15%	1239	13.54%	25.37%	Missingness not associated with elevated default. Variable remains difficult to interpret due to ambiguous units.

Missingness Assessment	Variables
Strong evidence of target leakage (≈100% default when missing)	Interest Rate, Interest Rate Spread, Property Value, LTV
Strong non-random missingness	Upfront Charges
Requires further investigation	DTI
Little evidence of missingness bias	Loan Limit, Income